COMPARATIVE STUDY ON THE EFFECTIVITY OF POWDERED CRAB SHELL AND EXTRACTED CHITIN TO THE GROWTH OF TOMATO PLANT

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Abstract

Utilization of crab shells is unheard of in a societal setting. Therefore, being surrounded with bodies of water, crab shells are one of the most generated waste products in Hagonoy, Bulacan, leading to unpleasant odors and poor environmental aesthetics. For this reason, a remediation method is necessary in order to alleviate this problem. Case in point is the utilization of powdered crab shell and extracted chitin as fertilizers as previous studies have proved it to be potent in enhancing the growth of plants due to its high nitrogen content and low C/N ratio. The preparation process involved physical and chemical treatments, such as the removal of protein and calcium carbonate with sodium hydroxide and hydrochloric acid, respectively. The products were then applied every week in varying amounts as fertilizer through soil application for a month-long duration. However, results showed that extracted chitin can be detrimental to plant growth if no regular monitoring of pH level take place. Hence, it causes the plant to raise its soil pH level and deteriorate. Consequently, an analysis of variance showed that although powdered crab shell enabled easier application as fertilizer, it had no significant effect on the growth of the tomato plant in terms of height, but is proven efficient in increasing the growth of leaves in terms of size. Which is why based on the conclusions of the study, the fertilizers' testing in other types of plants, constant pH level monitoring, and testing in different setting and condition is therefore recommended for future studies in relation to the topic under study.